Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/11/2025
PI Name:	Roberts, Donna M.D.		
Project Title:	Gender Difference and the Reversibility of the Structural Brain Changes of Spaceflight Correlated with Cognitive/Behavioral Performance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	07/01/2021	**End Date:**	09/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2023 per NSSC information (Ed., 7/13/2023). NOTE: End date changed to 06/30/2023 per L. Barnes-Moten/JSC (Ed., 4/22/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McTeague, Lisa Ph.D. (Medical University of South Carolina)		
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Performance Goal Text:			

Task Description:	We have previously shown an upward shift of the brain and crowding of eloquent brain tissue at the vertex in astronauts following long-duration missions aboard the International Space Station (ISS) and that long-duration, but not short-duration, spaceflight resulted in a significant increase in total ventricular volume (%?VV, 10.7% vs 0%, P<0.001). More recently, we found an association between the structural brain changes and altered cognitive and motor performance. In a small group of astronauts, we found that %?VV correlated significantly with worsening postural control (r=0.74, P<0.05, n=8), a decrement in accuracy for a learning-related code substitution test (r =-0.60, P<0.05, n=12) but faster reaction times for continuous performance testing (r =-0.62, P<0.05, n=12). Furthermore, we identified local brain structural changes post-flight which correlated with changes in postural control (left caudate), completion time of a complex motor task (right primary motor area/mid cingulate), and altered reaction times (bilateral optic radiations and splenium of the corpus callosum). We also found that astronauts who developed Spaceflight Associated Neuro-ocular Syndrome (SANS) demonstrated a significantly smaller change in %?VV than those without SANS (12.8% vs 6.5%, P=0.02, n=8 vs n=4). We hypothesize that these changes in brain structure contribute to altered cognitive and motor performance in astronauts post-flight. Limitations due to the retrospective nature of our study included the lack of long-term follow-up imaging and clinical assessments to determine whether or not the behavioral and structural brain changes were reversible. Another limitation was the small number of female astronauts. Here we propose to investigate gender differences and the reversibility of post-flight structural brain changes in astronauts after some time back in the 1 gravity environment of Earth and to assess for any association with altered cognitive/behavioral performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide insights into how alterations in brain structure may be correlated with performance changes and help identify anatomical structures involved in specific cognitive tasks. These findings may then be applied to specific patient populations experiencing cognitive deficits.
Task Progress:	In order to correlate cognitive performance with the structural brain changes noted in astronauts following space, a performance metric first needed to be developed. The WinSCAT (Windows Spaceflight Neuro-cognitive Assessment Tool) is a cognitive battery that has been routinely used to assess astronaut cognitive performance during International Space Station (ISS) missions. It is typically used by NASA as a monitoring device to flag large fluctuations in function, but a thorough analysis of changes in astronaut cognition using WinSCAT measures has not been published. The WinSCAT includes 5 cognitive tests and is administered in three phases: preflight (target 6 sessions); inflight (1 session every 30 days); and 1 session postflight. Therefore, in the initial months of this study, we developed a methodology for analyzing this longitudinal WinSCAT data. In addition, we also developed a pipeline for the analysis of longitudinal MRI (magnetic resonance imaging) data in astronauts which takes into account both global brain shift (not typically considered by standard terrestrial models) and local structural change. Based on Jacobian analysis, we have determined areas of the brain that demonstrate the greatest structural changes expressed as compression (yellow) vs. expansion (blue) at post-flight imaging compared with preflight imaging ($p < 0.05$). This is consistent with local expansion of the ventricles and compression of brain tissue and narrowing of the cerebrospinal fluid (CSF) spaces at the vertex. This year we completed correlational analysis between WinSCAT performance metrics and postflight structural brain changes on 41 astronauts. We found several brain regions which showed significant linear associations between regional means of the Jacobian determinant and WinSCAT throughput difference of slopes. Additionally, we have written up these results, presented them at the 2024 NASA Human Research Program Investigators' Workshop, and have prepared a manuscript for submission for peer review.
Bibliography Type:	Description: (Last Updated: 01/15/2026)